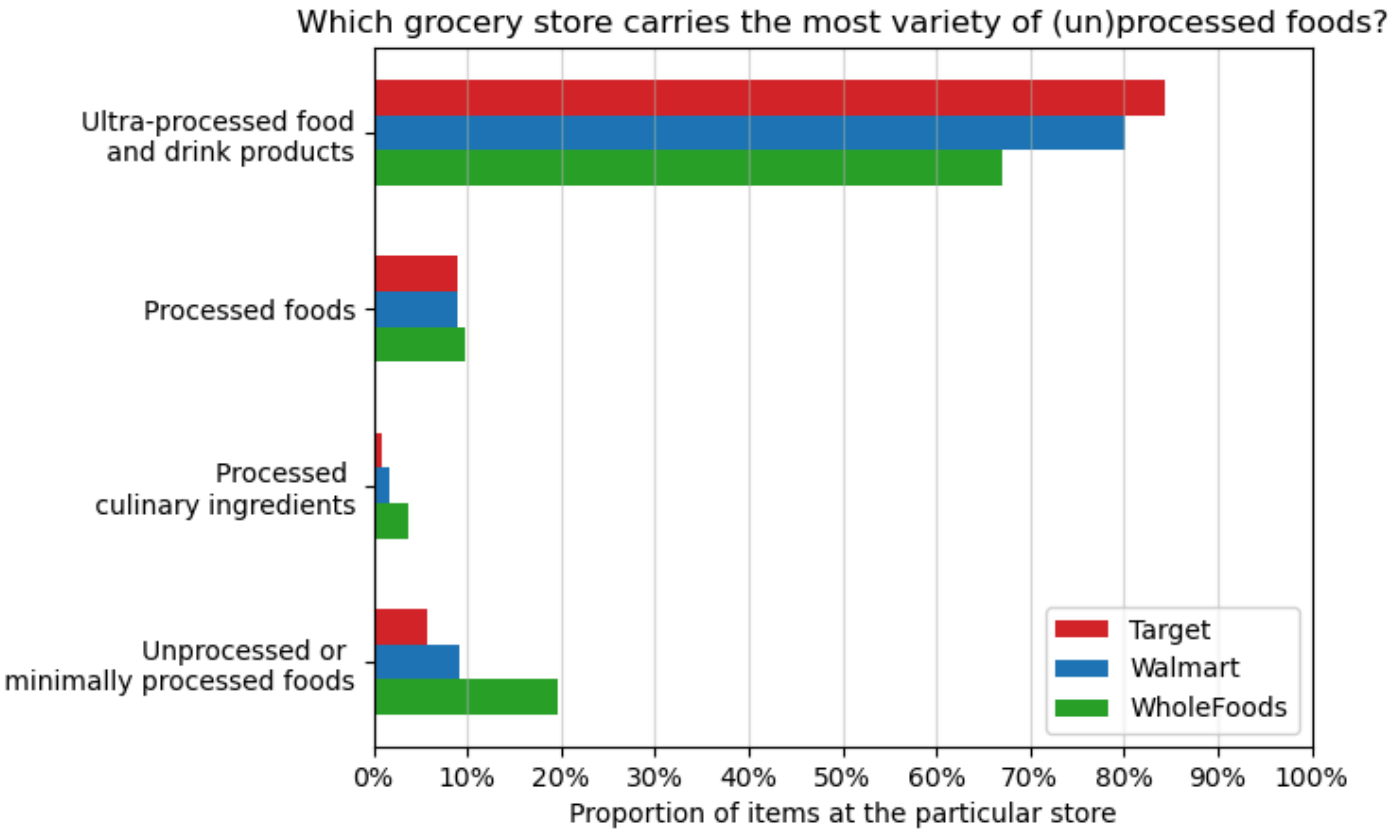

Which grocery store carries the most variety of (un)processed foods?
Ultra-processed food and drink products
Processed foods
Processed culinary ingredients
Unprocessed or minimally processed foods
Target
Walmart
WholeFoods
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
100%
Proportion of items at the particular store

When comparing Target, Walmart, and Whole Foods, many would point out their difference in quality and price. Whole Foods is generally perceived as higher quality, with more natural and organic ingredients, as opposed to Walmart, which gives the impression of cheap, processed food. The goal of this visualization is to prove (or disprove) that common perception using data, where every food product is labeled with a Food Processing level and a class.

Since the data attribute “food processing class” is categorical, I chose to present it using a bar chart. The classes are plotted against the proportion of items at the particular store. I transformed the item count into a ratio because a store can have more item entries than others, so this is a fairer way to compare the quantity of processed vs. unprocessed foods within the store instead of using raw counts. The bar chart is turned horizontally so that the class names can be easily read, which puts the ratio on the x-axis.

I avoided using separate bar plots for each store because area sizes are hard to compare when they are too far apart. Thus, my choices were overlapping bars, stacked bars, or grouped bars. Stacked bars are excluded because the baseline of each bar is not aligned across classes, which also makes it difficult to compare area sizes for each label (store). I tried to plot overlapping bars, and it was good at showing which store has the highest ratio for a class, but it was difficult to see the lower bars, even at low opacity. Thus, grouped bars were the best choice, where each store’s bar within the same class is placed side by side. The colors are chosen to match the grocery stores’ logos: Target is red, Walmart is blue, and Whole Foods is green. The order of the grouped bars also aligns with the order of labels in the legend.

Smaller details of the visualization include the gridlines and the % symbol for each x-axis tick. Gridlines are added for the x-axis because the top category bars are too far away from the x-axis, so grids help align the numbers and values of the bars. Although adding the % symbol after every x-tick might look cluttered, it helps the reader grasp the unit more quickly and makes it easier to understand the x-axis is item count ratio.